

Urbanization and Breast Cancer Incidence in North Carolina, 1995–1999

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PURPOSE: Breast cancer incidence rates are reported to be higher in urban compared with rural areas in the United States. We investigated the relationship between urbanization and breast cancer in North Carolina (1995–1999), and considered hospital characteristics as an explanation.

METHODS: We calculated age-adjusted *in situ* and invasive female breast cancer incidence rates stratified by race, urbanization (Urban Influence Codes), and the presence of a hospital with a cancer registry and cancer program approval in a county.

RESULTS: For white women, incidence rate ratios (IRRs) comparing the most urban with the most rural counties were 1.60 for *in situ* and 1.18 for invasive cancer. For non-white women, IRRs were 1.27 and 0.99, respectively. IRRs for incidence in registry hospital counties versus those without were all > 1.00 and differences were greater for *in situ* cancer than invasive. For most strata, urban excesses were attenuated when further stratified by registry hospital status.

CONCLUSIONS: For most strata, we observed excess incidence in urban counties, but it appeared to be explained through the urban preponderance of registry hospitals. Counties with these hospitals may have higher incidence because of increased detection. Area hospital characteristics should be considered when evaluating geographic patterns of breast cancer incidence.

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INTRODUCTION

Female breast cancer incidence rates have been reported to be higher in urban areas compared with rural areas in the United States since 1955. Past studies have found urban elevations with relative risks ranging from 1.06 to 1.83 (1–16). Only one estimate showed higher rates in rural areas, among African-Americans only (6). The urban excess has also been observed among males with breast cancer (17) and among females in Canada (18), European countries (17, 19–24), Costa Rica (25), and Japan (17).

Higher rates in urban areas such as Long Island, New York, have been used as a justification for studying environmental pollution as a cause of breast cancer (26).

Higher urban incidence may be the result of exposures, behaviors, or reproductive patterns more common in urban areas that elevate risk. However, higher rates in urban areas could arise from differential ascertainment due to higher rates of detection in urban areas, and consequently, higher reported incidence.

Quality of reporting has been considered as an explanation for the urban excess of breast cancer in six studies (1–3, 10–12). Most found that quality in rural areas was poorer by their indicator (percent of incident cases found by review of death certificates) (1–3, 10, 12). One study noted a potential reporting bias by “registry hospital,” noting that larger hospitals reported more completely, but did not conclude that this bias explained the urban excess (5). In a 1995 study, American hospitals with the largest cancer caseloads were found to more completely report their cases to the Surveillance, Epidemiology, and End Results (SEER) Program than hospitals with smaller cancer caseloads. Also, the percentage of cases missed and unreported to SEER was lower for hospital cancer registries compared with case-finding by the central cancer registry for the SEER region (27).

In North Carolina, a greater proportion of urban than rural hospitals are certified by the American College of Surgeons’ (ACOS) Commission on Cancer as “Approved Cancer Programs.” Such hospitals are required to have an in-house cancer registry (28). Our study was undertaken to investigate whether breast cancer incidence rates in urban

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Selected Abbreviations and Acronyms

SEER = Surveillance, Epidemiology, and End Results Program
ACOS = American College of Surgeons
NC = North Carolina
UIC = Urban Influence Code
IRR = incidence rate ratio

areas were higher than in rural areas in North Carolina (NC), measuring urbanization using a standardized, non-dichotomous scale that accounts for adjacency with Metropolitan Areas (29). We hypothesized that the presence of a hospital approved by the ACOS' Commission on Cancer in a county would also be related to higher incidence, possibly due to greater detection, more complete case-finding, or more complete case reporting.

MATERIALS AND METHODS

We conducted an ecologic study of female breast cancer incidence in the state of North Carolina, using county as the unit of analysis. Our study was approved by the Institutional Review Board of the School of Public Health at the University of North Carolina at Chapel Hill. We used incident case count data supplied by the North Carolina Central Cancer Registry; incident cases were North Carolina resident women with a first case of invasive or *in situ* breast cancer diagnosed between January 1995 and December 1999. For the purposes of this study, breast cancer incidence always refers to *detected* cases of breast cancer, not the true incidence of all cases.

Cancer registry data included age, race, county of patient's residence at diagnosis, year of diagnosis, and stage at diagnosis. There were 28,830 cases of breast cancer over the study years. We excluded 841 (2.9%) women with missing data on race (75), stage at diagnosis, (777), or both (11), leaving 27,989 cases for analysis. We stratified on race as the incidence of breast cancer is known to vary by race (30). Race was either self-reported or health-care-provider recorded; collection methods varied by county and facility. We used yearly population estimates provided by the State Demographer in order to calculate race and stage-specific incidence rates, averaged over the 5-year period. Rates were age-adjusted by the direct method to the year 2000 U.S. Standard Population and expressed per 100,000 women of the same race. Races were grouped as white and non-white, as reliable population denominators from the U.S. Census for smaller racial groups (in our data, American Indians and Asians) were not considered by the National Program of Cancer Registries to be reliable (31). We combined these groups with non-whites so that they could still contribute to the analysis although they accounted for less than 1% of all cases. Stage-specific rates were calculated as *in situ* and

invasive (defined as localized, regional, and distant stage combined).

Urbanization level was measured using the county's 1993 Urban Influence Code (UIC) based on the 1990 U.S. Census. This nine-level scale defines non-Metropolitan counties by their adjacency or non-adjacency to a Metropolitan Area, and the size of the largest communities within them (29). This scale is particularly useful for health research as the size of the largest town often defines whether hospitals are located within them, and adjacency may influence access to care (32). We used the full scale and a scale grouped as follows: 1–2 = Metropolitan Area; 3–6 = Non-Metropolitan, adjacent to a Metropolitan Area; and 7–9 = Non-Metropolitan, non-adjacent to a Metropolitan Area. Our rationale for this grouping was the assumption that among non-Metropolitan counties, non-adjacency to a Metropolitan Area may represent reduced availability of health care services for residents in these areas compared with non-Metropolitan, adjacent areas. The UIC definition takes into account Metropolitan Areas in adjacent states in their assignment of metropolitan status.

Rates were calculated for the entire state and for counties grouped by urbanization level. Incidence rate ratios (IRRs) were calculated comparing the most urban to the most rural group, and the most urban to the second most rural group. Because our study uses reported incidence and is not based on sampling of the NC population in the specified period, every reported case is (theoretically) captured. Thus, we did not calculate confidence intervals (which represent sampling variability) for our rates or proportions (33).

We also considered incidence rates by hospital type (having a hospital-based registry, or not), race, stage and grouped UIC. The presence of a hospital-based registry in a county was determined by whether or not any hospital(s) in a county was approved by the ACOS' Commission on Cancer in 2002 (28). Such hospitals were designated "registry hospitals" in this analysis, and each North Carolina county (N = 100) was assigned a yes/no variable indicating its presence. A county map was created that included the simultaneous geographic distribution of urbanization level (grouped UIC) and registry hospital status for each county (34). IRRs and incidence rate differences between registry and non-registry counties were calculated. We used the non-parametric Wilcoxon's rank sum to test the null hypothesis that registry and non-registry rates were a random partition of the same underlying distribution of rates.

RESULTS

Incidence Rates

For the state of North Carolina, the rate of average annual age-adjusted *in situ* breast cancer incidence for 1995 to 1999

was 22.5 per 100,000 for white women and 18.2 per 100,000 for non-white women. For invasive breast cancer, incidence rates were 118.2 and 104.6 for whites and non-whites, respectively. Considering rates by urbanization level, stage, and race, the highest rates among white women for *in situ* breast cancer were in the two most urban categories and the lowest rates were observed in the two most rural UICs (Table 1). Non-white women also had the highest *in situ* rates in the two most urban counties, but the lowest rates occurred in the middle of the scale. For invasive breast cancer, there was no consistent relationship with urbanization for either race across the full UIC scale.

Using rates aggregated over the grouped urbanization scale, *in situ* breast cancer incidence rates were highest in the most urbanized counties among whites (IRR = 1.60 comparing most urban with most rural) and among non-whites (IRR = 1.27 for the same comparison). Invasive breast cancer rates were higher in the most urban counties for whites (IRR = 1.18 comparing most urban with most rural) but not non-whites (IRR = 0.99 for the same comparison).

Registry Hospital Analysis

Registry hospitals were more likely to be located in Metropolitan counties (Fig. 1). As of 2002, there were 31 ACOS certified registry hospitals in North Carolina, and 21 (67.7%) were Metropolitan in location. Eight (25.8%) of the registry hospitals were in non-Metropolitan, adjacent

counties, and only two (6.5%) were in the most rural counties.

We observed higher *in situ* breast cancer incidence rates in counties with registry hospitals compared with counties without registry hospitals (Table 2). IRRs were 1.48 (comparing registry to non-registry county) for whites and 1.53 for non-whites, considering all of North Carolina. Patterns were similar for invasive breast cancer, although differences were of smaller magnitude than for *in situ* disease (IRR = 1.25 for whites and 1.18 for non-whites). For both *in situ* and invasive incidence in both racial groups, the excess persisted in analyses stratified by urbanization level (UIC groups).

Urban/Rural Differences—Explained by Registry Hospitals?

We then considered whether the urban excesses in incidence observed in most of our strata could be explained by the greater predominance of registry hospitals in urban areas. We observed that registry hospitals were related to both urbanization and higher incidence. Therefore, we recalculated our urban/rural incidence comparisons within strata of registry hospital status in order to observe if the relationship existed within the strata. The previously observed, overall most urban to most rural rate ratio is followed by the same comparison within strata of counties aggregated according to whether they had a registry hospital, or not (Table 3).

TABLE 1. Age-adjusted* average annual *in situ* and invasive breast cancer incidence rates per 100,000 women, 1995–1999, by race and Urban Influence Code of North Carolina county, N = 100

UIC	Whites	Invasive rate	Non-whites	Invasive rate
	<i>In situ</i> rate		<i>In situ</i> rate	
Most urban 1	26.2	116.6	20.3	91.0
2	23.9	124.8	20.8	111.9
3	18.6	105.5	14.2	96.1
4	18.0	125.6	12.8	107.1
5	20.9	124.2	11.4	98.1
6	20.3	109.2	15.7	97.3
7	21.2	118.6	18.5	119.2
8	13.7	99.6	13.5	100.8
Most rural 9	12.4	98.8	16.5	100.9
Grouped UICs	<i>In situ</i> rate (No. of cases)	Invasive rate (No. of cases)	<i>In situ</i> rate (No. of cases)	Invasive rate (No. of cases)
Metropolitan 1–2	24.5 (2,534)	122.7 (12,932)	20.7 (502)	107.3 (2,664)
Non-metropolitan adjacent 3–6	20.1 (854)	112.0 (4,793)	13.7 (173)	97.9 (1,250)
Non-metropolitan, non-adjacent 7–9	15.3 (241)	104.0 (1,623)	16.3 (56)	108.0 (367)
Rate ratio (1–2 vs. 7–9) [†]	1.60	1.18	1.27	0.99
Rate ratio (1–2 vs. 3–6) [‡]	1.22	1.10	1.51	1.10

*Age-adjusted to the year 2000 U.S. Standard Population.

[†]Rate ratios for the metropolitan (groups 1–2) versus the non-metropolitan, non-adjacent (groups 7–9).

[‡]Rate ratio for the metropolitan (groups 1–2) versus non-metropolitan adjacent (groups 3–6).

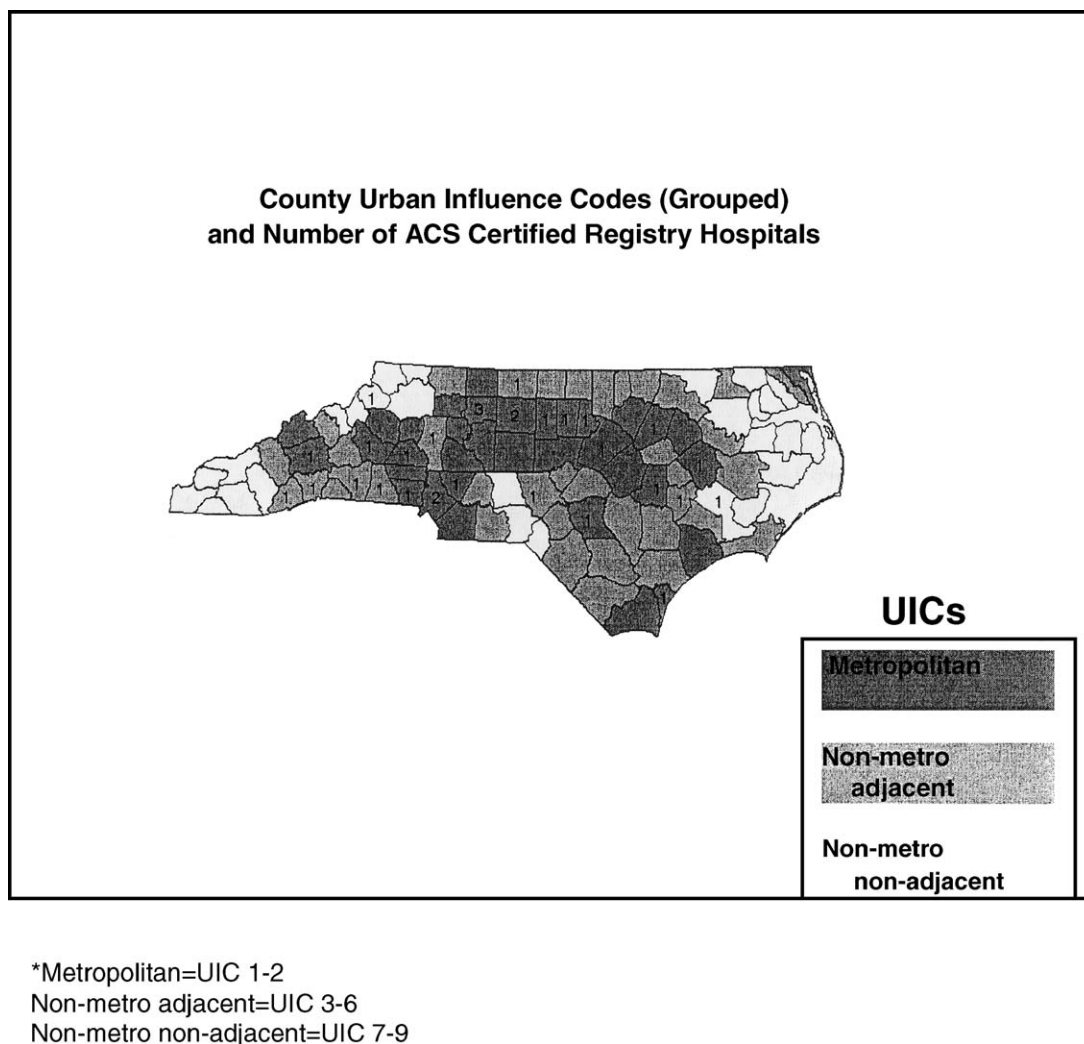


FIGURE 1. Distribution of grouped urban influence codes (UICs)* and number of American College of Surgeons certified registry hospitals, North Carolina, 2002.

For *in situ* breast cancer among whites, the previously observed urban/rural ratio (1.60) disappeared in counties with registry hospitals ($RR = 1.02$). Counties without registry hospitals still showed an urban excess (1.30). For non-white women, the previously observed urban excess ($IRR = 1.27$) was essentially null in the presence ($IRRs = 0.98$) and absence ($IRR = 1.00$) of a registry hospital, suggesting the presence of a registry hospital fully explained the urban-rural difference in non-white *in situ* rates.

For invasive breast cancer among whites, urban/rural rate ratios within strata of registry hospital were 1.07 and 1.02, as compared to the originally observed IRR of 1.18. Among non-whites, there had been no relationship between urbanization and invasive incidence (0.99), but the stratum-specific rate ratios showed inverse associations with urban

residence in the presence of a registry hospital (0.85) and in its absence (0.80). For these non-white women, registry hospitals appeared to mask higher *rural* rates of invasive disease.

DISCUSSION

We conducted an ecologic study to determine if an urban excess of breast cancer incidence was present in North Carolina for our study years, and, if so, to consider the role of hospital characteristics as a potential explanation. We found that greater urbanization, as represented by the grouped UIC, was related to higher *in situ* breast cancer incidence in white and non-white women and higher invasive incidence among white women but not non-white.

TABLE 2. Age-adjusted* average annual county-specific breast cancer *in situ* and invasive breast cancer incidence rates per 100,000 women, North Carolina, 1995–1999, and case counts, by race, presence of registry hospital and Urban Influence Code (grouped)

Group, urbanization level	With registry hospital rate (No. of cases)	Without rate (No. of cases)	Rate difference	Incidence rate ratio	Wilcoxon rank sum test p value
All counties	27 counties	73 counties			
<i>In situ</i>					
Whites	26.1 (2440)	17.6 (1189)	8.5	1.48	<0.0001
Non-whites	21.4 (490)	14.0 (241)	7.4	1.53	<0.01
Invasive					
Whites	129.2 (12271)	103.1 (7077)	26.1	1.25	<0.0001
Non-whites	111.6 (2618)	94.8 (1663)	16.8	1.18	0.04
Metropolitan counties	17 counties	18 counties			
<i>In situ</i>					
Whites	26.9 (2015)	18.0 (519)	8.9	1.49	<0.01
Non-whites	22.0 (435)	15.1 (67)	6.9	1.46	0.07
Invasive					
Whites	131.2 (9946)	102.1 (2986)	29.1	1.29	<0.001
Non-whites	112.5 (2280)	84.0 (384)	28.5	1.34	<0.01
Non-metropolitan adjacent	8 counties	31 counties			
<i>In situ</i>					
Whites	21.6 (357)	18.9 (497)	2.7	1.14	0.14
Non-whites	16.1 (42)	13.1 (131)	3.0	1.23	0.07
Invasive					
Whites	119.0 (1993)	105.9 (2800)	13.1	1.12	0.07
Non-whites	100.4 (266)	96.7 (984)	3.7	1.04	0.89
Non-metropolitan, non-adjacent	2 counties	24 counties			
<i>In situ</i>					
Whites	26.4 (68)	13.8 (173)	12.6	1.91	0.08
Non-whites	22.4 (13)	15.1 (43)	7.3	1.48	0.80
Invasive					
Whites	123.0 (332)	100.2 (1291)	22.8	1.23	0.18
Non-whites	132.2 (72)	104.7 (295)	27.5	1.26	0.08

*Age-adjusted to the year 2000 U.S. Standard Population.

Both invasive and *in situ* breast cancer incidence rates for both races were elevated among counties with registry hospitals, and with the exception of *in situ* disease among whites, urban elevations in rates were attenuated within strata defined by the presence of registry hospitals.

We are aware of no prior studies that consider breast cancer incidence by hospital characteristics. Variation in breast cancer incidence by urbanization level has not been previously studied in North Carolina, a state with a relatively large rural population (ranked sixth in the nation by the Census for proportion rural in 1990) (35). Prior ecologic studies of cancer incidence have varied with respect to the measurement of urbanization. Many American studies considered incidence as a function of population density, or have used a dichotomous definition of urban and rural areas (1–7, 9–11, 14, 36). Population density may perform poorly as a proxy for urbanization in large counties with a locus of population clustered around a single city (37), while dichotomous definitions may poorly capture heterogeneity within rural areas (32, 38). A recent study of cancer incidence using cancer registry data in Texas found residual

heterogeneity within non-Metropolitan Area rates when non-dichotomous urbanization scales such as the UIC or the similar Urban-Rural Continuum Code were applied (39).

Our use of a non-dichotomous measure also revealed unique racial patterns in incidence at the very rural end of the scale. Among non-white women, invasive incidence was highest in the most rural UIC groups compared with the other two groups. Also, although white incidence exceeded non-white incidence generally in this study, non-white incidence was higher than white incidence only in most rural UIC group, for both stages. Compared with urban non-whites, rural non-whites are of lower socioeconomic status in North Carolina (40). We hypothesize that these rural women may have especially benefited from federal funding programs for screening of low-income women in the 1990s, reflected in a “catch-up” in rural incidence rates (41).

We observed that incidence in counties with registry hospitals was higher than incidence in counties without registry hospitals for all of our strata, although some rates were based on small case counts. The presence of a registry hospital may be a marker for availability of health care

TABLE 3. Urban/rural* *in situ* and invasive breast cancer age-adjusted incidence rate[†] ratio comparisons within strata of county's registry hospital status, North Carolina, 1995–1999

Rate	Overall	With registry hospital	Without
<i>In situ</i> breast cancer			
Whites			
Most urban rate	24.5	26.9	18.0
Most rural rate	15.3	26.4	13.8
Rate ratio	1.60	1.02	1.30
Non-whites			
Most urban rate	20.7	22.0	15.1
Most rural rate	16.3	22.4	15.1
Rate ratio	1.27	0.98	1.00
Invasive breast cancer			
Whites			
Most urban rate	122.7	131.2	102.1
Most rural rate	104.0	123.0	100.2
Rate ratio	1.18	1.07	1.02
Non-whites			
Most urban rate	107.3	112.5	84.0
Most rural rate	108.0	132.2	104.7
Rate ratio	0.99	0.85	0.80

*Urban = Urban Influence Codes 1–2; rural = Urban Influence Codes 7–9.

[†]Age-adjusted to the year 2000 U.S. Standard Population.

services. We found that that physician density was also higher in urban counties and counties with a registry hospital (data not shown). Registry hospitals are required to have mammography available in-house or by referral, and mammography must meet the standards set by the American College of Radiology or the State Public Health Department (28). The magnitude of the differences between registry and non-registry hospital rates we observed was larger for *in situ* than invasive cancers. However, these hospitals may also ascertain and report cases more completely, so it is difficult to know if either or both explanations can account for the registry/non-registry incidence difference.

It is possible that the presence of staff dedicated to reporting cases in registry hospitals may result in more complete case-ascertainment and/or more complete reporting of cases ascertained than non-registry hospitals. For example, registry hospitals may be more likely to report cases that are patients of staff physicians, but diagnosed and treated outside the hospital setting. We cannot know this definitively, however, without access to proprietary data specifically containing results of studies of case finding and completeness of reporting in registry and non-registry hospitals.

There are other possible explanations for the excess incidence of breast cancer in registry hospital counties. Registry hospitals and their certification by the American College of Surgeons may attract more cancer patients for care and treatment. These patients may give the address of a friend or family member in the same county as the hospital

when traveling from rural counties, and this local, more urban, address becomes associated with the case and ultimately reflected in the incidence statistic. We are aware of no published studies of county of diagnosis of breast cancer versus patient's county of residence to inform this hypothesis. One unpublished analysis completed in the early 1990s of women in 39 North Carolina counties in the central and eastern part of the state (that did not share a border with another state) found that 26% of women with breast cancer were diagnosed outside their county of residence. In addition, the six counties that received the highest proportion of cases from outside their county were located in Metropolitan Areas (42). This supports the idea that some women may travel from a non-Metropolitan county to a Metropolitan county for diagnosis. The expansion of ambulatory surgery centers in North Carolina in the 1990s, however, has facilitated outpatient treatment and diagnosis of breast cancer at a more local level, possibly reducing the need for travel. Due to patient confidentiality we have no individual-level data in this analysis that would allow us to consider the accuracy of patient address.

Women in counties with registry hospitals may have higher breast cancer risk due to a different distribution of risk factors in those counties. For example, women living near major urban academic centers such as Raleigh-Durham-Chapel Hill, the "Research Triangle," may have lower parity and later age at first full-term pregnancy than urban women elsewhere in North Carolina. We considered a county-level variable estimating parity (the mean number of children ever born to women aged 35–44 years) and found it was lower in registry hospital counties (1.8) versus counties without (2.0), suggesting that risk factors may indeed differ (43). This relationship held within grouped urbanization level. We also considered an ecologic measure of socioeconomic status, but did not find that the proportion of persons living in poverty varied consistently by urbanization level and registry hospital status (44). It is unknown which of the possible explanations presented above may account for our observations; it is possible that all are functioning simultaneously making it difficult to isolate the relative contribution of a single explanation. There may be additional unmeasured variables that are not captured by our study. Among whites for *in situ* disease, an excess incidence associated with the most urban counties persisted after stratification by hospital type, and remains unexplained in our analysis.

A limitation of these analyses is that the registry status of each hospital is known only for 2002; exact certification dates are held in confidence by the ACOS, so misclassification of registry hospital status over the study period is possible. Over the study period, urbanization has increased in North Carolina by population in-migration to Metropolitan Areas (45). The most recently available 1993 UICs

are based on the 1990 Census, so the degree of urbanization in our study may be inaccurate. If rural areas were more urban than we have measured, then the differences we observed would be attenuated. However, most of the population gain over this period was in areas already classified as urban, so such migration should not greatly change our estimates (45).

Our results call for further examination of the role of the health care availability when evaluating geographic patterns of breast cancer incidence. The explanatory factors behind higher incidence in the presence of a registry hospital are unknown, but could include more complete reporting due to the presence of the in-house cancer registry and differentially greater detection due to mammographic screening. Closer study of the exact mechanism(s) by which registry hospitals increase incidence is warranted. We suggest that urban excesses in reported breast cancer in other states may be explained by simple disparities in health care availability, a readily testable hypothesis. This approach may be a cost-effective first step towards understanding geographic patterns of incidence before embarking on more costly biologic exposure studies.

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